

JASON DHAKI

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PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate (CGPA 4.00/4.00) who ships production full-stack web, mobile, and AI applications — React and Next.js frontends, Node.js and Python FastAPI backends, PostgreSQL data layers, and LLM-powered agent systems built with Retrieval-Augmented Generation (RAG), LangChain/LangGraph, and vector databases. 28 public repositories spanning full-stack platforms, AI agents, Python automation, and embedded robotics. Seeking software engineering, full-stack, mobile, and AI/automation roles.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL, HTML5, CSS3

Frontend: React.js, Next.js, React Native, Tailwind CSS, DaisyUI, NativeWind, Framer Motion, Three.js, Responsive Web Design

Backend & APIs: Node.js, Express.js, Python FastAPI, REST APIs, Streamlit, Clerk, NextAuth, Stripe, SSLCommerz, Twilio

AI & Automation: Large Language Model (LLM) APIs (Anthropic Claude, Groq, Gemini, Ollama), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, AI Agents, Prompt Engineering, ChromaDB, pgvector, EasyOCR, Python Automation Scripting

Databases: PostgreSQL (Neon), MongoDB, SQLite, Redis, Neo4j, Prisma ORM, SQLAlchemy, Alembic, Sanity Headless CMS, GROQ

Cloud, DevOps & Testing: Git, GitHub, Docker, Vercel, Render, Heroku, Backblaze B2 (S3-compatible storage), Virtual Private Server (VPS) deployment, pytest

Mobile Development: React Native, Expo, Expo Router, Gradle, Android SDK CLI Tools, cross-platform builds

Hardware & Embedded Systems: Arduino (MEGA/UNO), Robotics, ROS, L298N Motor Drivers, QTR-8A Sensor Arrays, Ultrasonic Sensors, AutoCAD, PSpice/OrCAD

CS Fundamentals & Practices: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Systems, Operating Systems, Software Architecture, API Design, Relational Data Modeling, Unit Testing, Agile Collaboration

PROJECTS

AI5K Profile Intelligence — Evidence-Grounded Profile Analyzer | *Python, FastAPI, PostgreSQL/pgvector, SQLAlchemy, Alembic, Docker, Render*

- Architected a modular-monolith FastAPI service that ingests CVs, GitHub profiles, and freelance profile text, extracts claims through parallel LLM calls, and re-slices each claim to a literal source span so no generated statement appears without a traceable citation.
- Built a 7-dimension scoring engine with versioned benchmarks, rules-based evidence tiering, and a gap list ranked by points-per-hour to surface the highest-return improvements.
- Designed swappable backends behind clean interfaces — Neon PostgreSQL with pgvector via SQLAlchemy and Alembic, local disk or Backblaze B2 object storage, and Anthropic or Groq as LLM provider — each switchable by environment variable.
- Containerized with Docker and deployed to Render with automated pre-deploy migrations, backed by 19 pytest tests covering span round-trips and schema invariants.

EWU Policy Expert — Agentic RAG Assistant | *Python, LangChain, LangGraph, ChromaDB, Groq (Llama 3.3 70B), Streamlit*

- Built and deployed a live agentic RAG assistant that answers university disciplinary and examination policy questions grounded strictly in ingested PDF policy documents.
- Orchestrated a LangGraph workflow that combines ChromaDB vector retrieval with a deterministic Python severity-scoring function (0–10) before final answer synthesis.
- Migrated inference from local Llama 3 (Ollama) to the Groq Cloud API (Llama-3.3-70b) for faster responses, and solved Streamlit Cloud's ephemeral filesystem limits with a Git-backed persistent vector database.

TrustLens AI — Bilingual Misinformation Detection Platform | *Next.js, Python, FastAPI, Redis, Neo4j* — *Infinity AI BuildFest 2026 Hackathon* | Live: trust-lens-ai-beta.vercel.app

- Built a bilingual trust-scoring system that rates any claim from 0–100 across six explainable pillars, delivered through a Next.js web app, a Telegram bot, and a Chrome extension.
- Developed and deployed high-throughput FastAPI endpoints on a virtual private server with Redis caching and a Neo4j graph database to detect misinformation across the Bangladeshi internet ecosystem in real time.

Jhunu's Craft — Full-Stack B2B/B2C Export Platform | *Next.js, TypeScript, Prisma, PostgreSQL, Stripe, NextAuth* | Live: jhunus-craft.vercel.app

- Architected a full-stack e-commerce and B2B wholesale export platform for sustainable Bangladeshi jute products, streamlining the international supply chain from catalog to checkout.

- Designed the relational database schema with Prisma ORM and PostgreSQL to handle wholesale orders, secure B2B authentication via NextAuth, and Stripe-based global transactions.

Hope Crafts — Headless-CMS Commerce Platform | *Next.js, TypeScript, Sanity CMS, GROQ, Clerk, Stripe, SSLCommerz* | Live: hopescraft-cvmz.vercel.app

- Developed a heritage-driven e-commerce and storytelling platform supporting 500+ artisans, built on a headless Sanity CMS architecture with optimized GROQ queries for fast content delivery.
- Integrated multi-gateway checkout — Stripe for international buyers and SSLCommerz for local payments — alongside Clerk authentication and Vercel deployment.

Gym Levelling — Gamified Fitness Mobile Application | *React Native, Expo, Expo Router, TypeScript, NativeWind, Gradle, Android SDK*

- Developed a cross-platform mobile MVP that maps workout consistency to RPG-style character-attribute progression to improve training adherence.
- Configured native Android builds with Gradle and Android SDK CLI tools, using Expo Router for navigation and NativeWind for styling.

Dukan Dost — Voice and OCR Inventory Assistant | *Python, Twilio, Google Speech-to-Text, EasyOCR, SQL, Replit/Heroku* — Hackathon, 2-person team

- Co-built an inventory assistant for small shopkeepers that accepts spoken input via the Google Speech API and extracts stock data from receipt images with an EasyOCR pipeline.
- Implemented the inventory database layer and Twilio messaging integration, then packaged the service for cloud deployment.

Autonomous Line-Following Robot with 4-DOF Robotic Arm | *Arduino MEGA, Embedded C/C++, QTR-8A Sensor Array, L298N Motor Driver, Ultrasonic Sensor*

- Engineered an autonomous line-following and obstacle-avoidance vehicle using a QTR-8A reflectance sensor array, ultrasonic distance sensing, and L298N motor drivers.
- Programmed a 4-DOF robotic arm for object pickup and tuned sensor calibration and motor-control logic for reliable track completion.

Additional Projects | *selected from 28 public repositories*

- Developer Portfolio (Next.js, Three.js, Framer Motion): performance-tuned personal site with a live WebGL hero, scroll-triggered reveals, and an SEO-ready foundation.
- French Learning Web App (React, Vite, TypeScript): CEFR A1–A2 study tool with exercises, quizzes, and an SM-2 spaced-repetition vocabulary review engine.
- Desktop Automation Suite (Python): utility scripts that organize Downloads by file type, flatten nested archive folders, and prepare a workstation for focused work.
- Core Systems Projects (Java, C/C++): OOP payroll model with encapsulation, inheritance, and polymorphism; school database management system; vehicle management system; password strength analyzer; login portal.

EXPERIENCE

A Levels Economics Lecturer — BACHA English Medium School

Sep 2024 – Present

- Design and deliver structured lesson plans covering macroeconomic and microeconomic theory, translating complex models into digestible concepts for GCE A-Level candidates.
- Grade examination scripts and provide targeted written feedback, tracking recurring weak points to adjust subsequent lessons.

Private Tutor — Independent Practice

2022 – 2026

- Delivered personalized instruction to students from Standard 6 through GCE A-Levels, independently managing scheduling, curriculum planning, and progress reporting across a rolling roster.

Volunteer Coordinator — Purnota Foundation

2024 – Present

- Direct volunteer teams and logistics for Eid-ul-Fitr ration distribution in Mohammadpur, ensuring safe and efficient operations during high-pressure events.

EDUCATION

Bachelor of Science in Computer Science and Engineering

Expected Jan 2028

East West University, Dhaka, Bangladesh | CGPA: 4.00/4.00 | Dean's Scholarship

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Operating Systems, Electronic Circuits, Computer-Aided Drawing (AutoCAD), Business Communications

CERTIFICATIONS & ACHIEVEMENTS

Languages: IELTS 8.0/9.0 (C1 English Proficiency) · DELF A2 (French) · Bengali (Native)

Awards: Dean's Scholarship · The Daily Star Award 2026 & 2023 (GCE A & O Levels) · BACHA Day Coordinator Award 2024